

2(a)	(electric) force is (directly) proportional to product of charges	B1
	force (between point charges) is inversely proportional to the square of their separation	B1
2(b)(i)	(electric) force is perpendicular to velocity (of particles)	B1
	force (perpendicular to velocity) causes centripetal acceleration or force does not change the speed of the particles or force has constant magnitude	B1
2(b)(ii)	$F = e^2 / 4\pi\epsilon_0 x^2$	C1
	$= (1.60 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (2 \times 1.59 \times 10^{-10})^2]$	A1
	$= 2.28 \times 10^{-9} \text{ N}$	
2(b)(iii)	$F = mr\omega^2$ and $\omega = 2\pi / T$ or $F = mv^2 / r$ and $v = 2\pi r / T$	C1
	$F = 4\pi^2 mr / T^2$	C1
	$T = \sqrt{[4\pi^2 \times 9.11 \times 10^{-31} \times 1.59 \times 10^{-10} / (2.28 \times 10^{-9})]}$	
	$= 1.58 \times 10^{-15} \text{ s}$	A1
2(c)(i)	- electron and positron interact - positron is anti-particle of electron (pair) annihilation occurs <i>Any two points, 1 mark each</i>	B2
	mass of the electron and positron converted into photon energy	B1
2(c)(ii)	PET scanning	B1